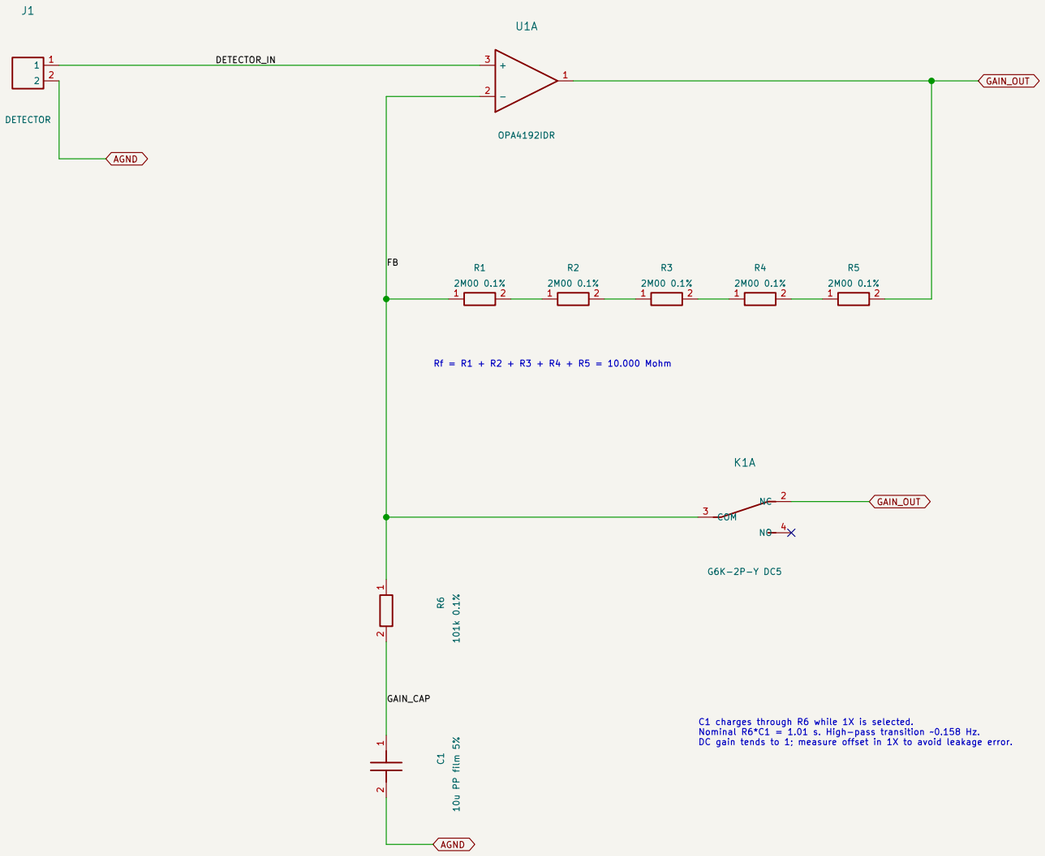
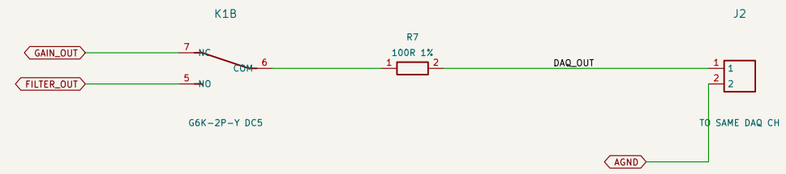
1X OFFSET + SENSITIVITY / 100X FILTERED NOISE

One OPA4192 section replaces one existing follower. Same detector bias/loading, +5 V supply and DAQ channel.

K1 is drawn UNPOWERED: gain resistor bypassed; unfiltered follower output selected.





MODE	K1	SIGNAL TO DAQ
BUFFER / SENSITIVITY	OFF	-1X, direct follower
NOISE	ON	-100X AC, 4-pole low-pass

0.7 V offset + 500 mV p-p -> about 0.45 to 0.95 V at 1X.
Never run sensitivity illumination in NOISE mode.
Relay contacts move at different times: discard all transitions.

Switching: stimulus OFF -> settle in 1X -> select NOISE -> wait for electrical/thermal settling -> record noise. Return to 1X before restoring the sensitivity stimulus.

Two unity-gain low-pass sections

File: noise_filter.kicad_sch

Power, manual/DAQ mode control

File: power_control.kicad_sch

SCHEMATIC ONLY — NOT A FABRICATION RELEASE

Eltec — engineering reference

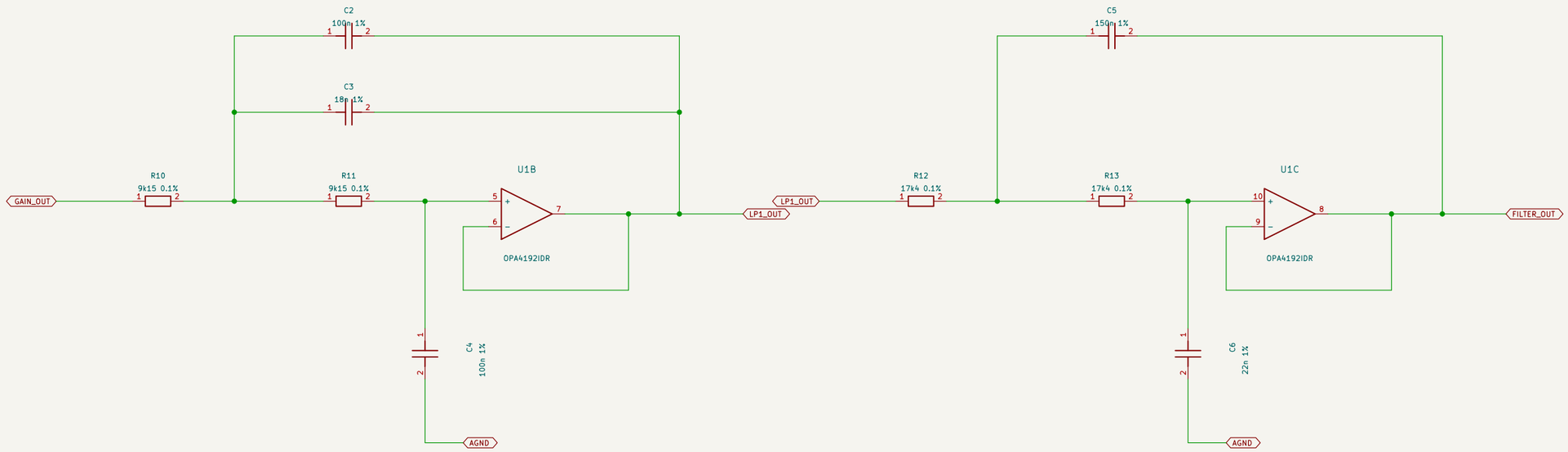
Sheet: /
File: switchable_gain.kicad_sch

Title: 1X / 100X switching amplifier

Size: A3	Date: 2026-09-14	**Rev: R1 PROTO**
KiCad E.D.A. 10.0.6		Id: 1/3

HARDWARE LOW-PASS: PASSES THE NOISE SIGNAL THROUGH 100 Hz

Two unity-gain Sallen-Key sections. Lower-Q section first; filter is bypassed at the DAQ in 1X mode.



SECTION 1: R10=R11=9.15k, Cf=C2+C3=118n, Cg=C4=100n
f0 = 160.125 Hz; Q = 0.54314
SECTION 2: R12=R13=17.4k, Cf=C5=150n, Cg=C6=22n
f0 = 159.226 Hz; Q = 1.30558

Nominal response (ideal amplifiers, before 100R/DAQ loading):
3 Hz: 99.87X 50 Hz: 100.18X 100 Hz: 99.35X
200 Hz: 37.65X 500 Hz: 1.04X 1 kHz: 0.065X
Values approximate a fourth-order Butterworth response.

Build requirements:
- No X7R/Y5V in the gain or filter signal capacitors.
- Keep C2/C3 and C5 return paths at their own op-amp outputs.
- Keep detector/feedback wiring away from relay coil and digital returns.
- Confirm capacitive-load stability with the actual DAQ cable.
- Calibrate each channel; calculated response is not a noise-floor test.

For 50 channels: 150 active amplifier sections = 38 OPA4192 quads total. Existing 13 quads can supply the 50 gain stages; 25 more quads supply the 100 filter stages. This drawing is a one-channel reference, not a completed 50-channel PCB or harness drawing.

SCHEMATIC ONLY — NOT A FABRICATION RELEASE
Eltec — engineering reference

Sheet: /Two unity-gain low-pass sections/		
File: noise_filter.kicad_sch		
Title: 100X noise filter / 160 Hz low-pass		
Size: A3	Date: 2026-09-14	**Rev: R1 PROTO**
KiCad E.D.A. 10.0.6		Id: 2/3

REGULATED ANALOG SUPPLY / SEPARATE RELAY SUPPLY / MODE CONTROL

Prototype uses one complete OPA4192. Units A–C on sheets 1–2; unit D parked below.

J3

REGULATED POWER

U1E

OPA4192IDR

C10 100n X7R

C11 10u X7R

C12 100n X7R

C13 10u X7R

R30 0R STAR

Route coil return separately to this star.
The existing LT1763 analog rail must NOT power 50 relay coils.

K1C

G6K-2F-Y DC5

D1 1N4148W

Q1 AO3400A

R21 1k 1%

R22 100k 1%

JP1

SW1 OPEN=1X / CLOSED=NOISE

R20 100k 1%

J4

DAQ CONTROL (OPTION)

Manual default: JP1 shunt 1–2; SW1 OPEN. No shunt also de-energizes K1.
AUTO: initialize DAQ DIO LOW before fitting 2–3 / enabling relay power.
DAQ DIO lines may power up pulled HIGH. Auto default is NOT guaranteed
by R22 while a connected DAQ actively pulls/drives the command HIGH.

R31 30k 1%

R32 10k 1%

U1D

OPA4192IDR

K1 coil: about 21.1 mA at 5 V per channel.
50 coils: ~1.06 A nominal, ~1.24 A at +5% rail / −10% resistance.
Use a separate regulated 5 V, >=1.5 A relay supply for 50 channels.
Analog: ~150 mA typical for 150 active sections; budget full quads.
No coil PWM; keep relay heat and coil wiring away from R1–R6/C1.

SCHEMATIC ONLY — NOT A FABRICATION RELEASE

Eltec — engineering reference

Sheet: /Power, manual/DAQ mode control/
File: power_control.kicad_sch

Title: Power and relay control / manual default

Size: A3	Date: 2026-09-14	**Rev: R1 PROTO**
KiCad E.D.A. 10.0.6		Id: 3/3